Exercise #5

13. February 2023

Problem 1. (Trigonometric series)

Determine the coefficients $a_0, a_n, b_n,\ n \in \mathbb{N}$ of the trigonometric Fourier series $f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(\frac{n\pi}{L}x) + b_n \sin(\frac{n\pi}{L}x)$ for the follwing functions

a) $f(x) = 1/2 + \cos(2x) - 4\sin(4x)$, (find the fundamental period yourself);

b) $f(x) = 1 + \sin^2(x)$, which has period $T = 2L = \pi$;

c) $f(x) = 1 - x$ for $-1 \le x \le 1$, with period $T = 2L = 2$;

d) $f(x) = |\cos(3\pi x)|$, that has period $T = 2L = 1/3$.

Solution.

a) We can recognise directly that $\cos(2x)$ has period $T = \pi$ and $\sin(4x)$ has $T = \pi/2$. The overall period is thus $T = 2L = \pi$ and the Fourier expansion $f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(2nx) +$

$b_n \sin(2nx)$. By the definition of the coefficients we have:

$$\begin{aligned}
a_0 =& \frac{1}{2\pi}\int_{-\pi}^{\pi}(1/2+\cos(2x)-4\sin(4x))\mathrm{d}x\\
=& \frac{\pi+\pi}{4\pi}+\frac{1}{4\pi}[\sin(2x)]_{-\pi}^{\pi}-\frac{1}{2\pi}[-\cos(4x)]_{-\pi}^{\pi}=\frac{1}{2};\\
a_n =& \frac{1}{\pi}\int_{-\pi}^{\pi}(1/2+\cos(2x)-4\sin(4x))\cos(2nx)\mathrm{d}x\\
=& \frac{1}{4n\pi}[\sin(2nx)]_{-\pi}^{\pi}+\frac{1}{\pi}\int_{-\pi}^{\pi}\cos(2x)\cos(2nx)\mathrm{d}x-\frac{4}{\pi}\int_{-\pi}^{\pi}\sin(4x)\cos(2nx)\mathrm{d}x;\\
b_n =& \frac{1}{\pi}\int_{-\pi}^{\pi}(1/2+\cos(2x)-4\sin(4x))\sin(2nx)\mathrm{d}x\\
=& \frac{1}{4n\pi}[-\cos(2nx)]_{-\pi}^{\pi}+\frac{1}{\pi}\int_{-\pi}^{\pi}\cos(2x)\sin(2nx)\mathrm{d}x-\frac{4}{\pi}\int_{-\pi}^{\pi}\sin(4x)\sin(2nx)\mathrm{d}x.
\end{aligned} \tag{1}$$

Since $\int_{-\pi}^{\pi}\sin(mx)\cos(nx) = 0, \ \forall m,n, \ \int_{-\pi}^{\pi}\cos(mx)\cos(nx) = 0, \ \text{ if } m \neq n$ and $\int_{-\pi}^{\pi}\sin(mx)\sin(nx) = 0, \ \text{ if } m \neq n$, we remain with:

$$\begin{aligned}
a_1 =& \frac{1}{\pi}\int_{-\pi}^{\pi}\cos(2x)^2\mathrm{d}x = \frac{1}{\pi}\int_{-\pi}^{\pi}\frac{1+\cos(4x)}{2}\mathrm{d}x = 1\\
b_2 =& -\frac{4}{\pi}\int_{-\pi}^{\pi}\sin(4x)^2\mathrm{d}x = -\frac{4}{\pi}\int_{-\pi}^{\pi}\frac{1-\cos(8x)}{2}\mathrm{d}x = -4
\end{aligned} \tag{2}$$

Direct comparison with the general formula, recognising in $f(x)$ terms of the Fourier series itself, gives directly: $a_0 = \frac{1}{2}, \ a_1 = 1, \ b_2 = -4$ in accordance to what found.

b) We see that $f(x)$ is an even function with period π, which means that we should find the coefficients to a Fourier seris of the form

$$a_0 + a_1\cos(2x) + a_2\cos(4x) + a_3\cos(6x) + \dots \tag{3}$$

Alternative 1:
We see that by trigonometric identities $f(x)$ can be written as

$$f(x) = 1 + \sin^2(x) = \frac{3}{2} - \frac{1}{2}\cos(2x). \tag{4}$$

Since $f(x)$ is already written on the form of a trigonometric Fourier series, we can see that

$$a_0 = \frac{3}{2}, \quad a_1 = -\frac{1}{2},$$

and all the other coefficients are 0.
Alternative 2:
Since $f(x)$ is even we have that $b_n = 0$. Calculating a_0 and a_n from the following integrals

$$a_0 = \frac{2}{\pi}\int_0^{\pi/2} (1+\sin^2(x))\mathrm{d}x,$$
$$a_n = \frac{4}{\pi}\int_0^{\pi/2} (1+\sin^2(x))\cos(2nx)\mathrm{d}x,$$

by using (4) when integrating, gives the same answer as alternative 1.

c) Given the Fourier series for this period as $f \sim a_0 + \sum_{n=1}^{\infty} a_n\cos(n\pi x) + b_n\sin(n\pi x)$, I thus just need to find b_n for f. This is because $a_0 = 1$ is just a constant and $-x$ is an odd function. I have:

$$\begin{aligned} b_n &= -2\int_0^1 x\sin(n\pi x)\mathrm{d}x \\ &= -\frac{2}{n\pi}\left[-x\cos(n\pi x)\right]_0^1 - \frac{2}{n\pi}\int_0^1 \cos(n\pi x)\mathrm{d}x \\ &= \frac{2\cos(n\pi)}{n\pi} - \frac{2}{n^2\pi^2}\left[\sin(n\pi x)\right]_0^1 \\ &= \frac{2(-1)^n}{n\pi} \end{aligned}$$

together with $a_0 = 1$, which is given directly.

d) We see that $f(x)$ is a even function, its Fourier series is $f \sim a_0 + \sum_{n=1}^{\infty} b_n\sin(6n\pi x)$. Furthermore $|\cos(3\pi x)| = \cos(3\pi x)$ in the interval considered. Calculating a_0, using that the function is even:

$$a_0 = 6\int_0^{1/6}\cos(3\pi x)\mathrm{d}x = \frac{2}{\pi}\left(\sin\left(\frac{\pi}{2}\right) - \sin(0)\right)\mathrm{d}x = \frac{2}{\pi}$$

Calculating a_n, using that $2\cos(\alpha)\cos(\beta) = \cos(\alpha+\beta)+\cos(\alpha-\beta)$

$$
\begin{aligned}
a_n &= 12\int_0^{1/6} \cos(3\pi x)\cos(6n\pi x)\mathrm{d}x \\
&= 6\int_0^{1/6} \left(\cos\left((1+2n)3\pi x\right) + \cos\left((1-2n)3\pi x\right)\right)\mathrm{d}x, \\
&= 6\left(\frac{1}{3\pi(1+2n)}\left(\sin\left((1+2n)\frac{\pi}{2}\right) - \sin(0)\right) + \frac{1}{3\pi(1-2n)}\left(\sin\left((1-2n)\frac{\pi}{2}\right) - \sin(0)\right)\right) \\
&= \frac{2}{\pi}\left(\frac{(-1)^n}{1+2n} + \frac{(-1)^n}{1-2n}\right) \\
&= \frac{4(-1)^n}{\pi(1-4n^2)}
\end{aligned}
$$

Between the third and fourth line above we have used that

$$\sin\left((1+2n)\frac{\pi}{2}\right) = (-1)^n.$$

Problem 2. (Complex series)

Determine the coefficients c_k of the complex Fourier series $f \sim \sum_{k=-\infty}^{\infty} c_k e^{i\frac{k n\pi}{L}x}$ for the following functions

a) $f(x) = -3e^{-4ix} + e^{-ix} + 4e^{3ix}$, (indentify the fundamental period yourself),

b) $f(x) = \cos^2(x)$, for $-\pi/2 \le x \le \pi/2$, with period $T = 2L = \pi$,

c) $f(x) = x^2 - x$ for $-1 \le x \le 1$, with period $T = 2L = 2$,

d) $f(x) = \mathrm{e}^{-x}$ for $-1 \le x \le 1$, with period $T = 2L = 2$,

Solution.

a) We start by noting that: $e^{-4ix} = \cos(4x) - i\sin(4x)$, which has period $T = 2L = \pi/2$. Same applies with e^{-ix} and e^{3ix} with periods $T = 2\pi$ and $T = \frac{2}{3}\pi$ respectively. The overall period is thus 2π and the complex Fourier series is $f \sim \sum_{k=-\infty}^{\infty} c_k e^{iknx}$. We can thus deduct, with the comparison between the latter and f that: $c_{-4} = -3,\ c_{-1} = 1,\ c_3 = 4$.

b) We first rewrite $f(x)$ to be expressed by complex functions,

$$f(x) = \cos^2(x) = \frac{1 + \cos(2x)}{2} = \frac{2 + e^{2ix} + e^{-2ix}}{4},$$

and note that the period is π, thus $f \sim \sum_{k=-\infty}^{\infty} c_k e^{i2kx}$. We now have the function on the form of a complex Fourier series, and can see that

$$c_0 = \frac{1}{2}, \quad c_1 = c_{-1} = \frac{1}{4}.$$

Alternatively, the coefficients can be found by integrating.

c)

$$c_k = \frac{1}{2}\int_{-1}^{1} \mathrm{e}^{-x}\mathrm{e}^{-ik\pi x}\,\mathrm{d}x = \left[-\frac{\mathrm{e}^{-(1+ik\pi)x}}{2(1+ik\pi)}\right]_{-1}^{1} = \frac{e^{1+ik\pi} - e^{-(1+ik\pi)}}{2(1+ik\pi)} = \frac{\sinh(1)}{1+ik\pi}(-1)^k.$$

Problem 3. (Parseval's theorem)

Consider two periodic functions $f(x)$ and $g(x)$ and their truncated Fourier series $f_N(x)$ and $g_N(x)$. The errors E_N between each function and its trigonometric approximation can be computed using Parseval's identity.

a) What does the formula for E_N looks like?

b) Let $f(x) = \mathrm{e}^{-x}$ for $-1 \le x \le 1$, with period $T = 2L = 2$. Based on the Fourier coefficients of $f(x)$, calculate the error E_N for $N = 2, 4$ and 8.
Hint: you have already computed the (complex) coefficients for this function on for Problem 2 (d).

c) Now, do the same for $g(x) = \mathrm{e}^{-|x|},\ -1 \le x \le 1$, with period $T = 2L = 2$.

d) Why is the error so much larger when approximating $f(x)$, than in the case of $g(x)$?

Solution.

a) From Parseval's identity, we have that

$$E_N = \int_{-1}^{1} f^2(x)\,\mathrm{d}x - 2\sum_{k=-N}^{N} |c_k|^2 .$$

b) We first integrate

$$\int_{-1}^{1} f^2(x)\mathrm{d}x = \int_{-1}^{1} e^{-2x}\mathrm{d}x = \frac{e^2 - e^{-2}}{2} = \sinh(2)$$

$$E_N = \sinh(2) - 2\sum_{k=-N}^{N} \frac{\sinh^2(1)}{(1+ik\pi)^2} = \sinh(2) - \sum_{k=-N}^{N} \frac{2\sinh^2(1)}{1+k^2\pi^2}$$

Inserting values for N: $E_2 = 0.22, E_4 = 0.123, E_8 = 0.066$

c) We first find the Fourier coefficients for $g(x)$

$$\begin{aligned}
c_k &= \frac{1}{2}\int_{-1}^{1} \mathrm{e}^{-|x|}\mathrm{e}^{-ik\pi x}\,\mathrm{d}x = \frac{1}{2}\int_{-1}^{0} \mathrm{e}^{x}\mathrm{e}^{-ik\pi x}\,\mathrm{d}x + \frac{1}{2}\int_{0}^{1} \mathrm{e}^{-x}\mathrm{e}^{-ik\pi x}\,\mathrm{d}x \\
&= \frac{1}{2}\left(\frac{1}{1-ik\pi}\left(1 - e^{-(1-ik\pi)}\right) - \frac{1}{1+ik\pi}\left(e^{-(1+ik\pi)} - 1\right)\right) \\
&= \frac{1}{2}\frac{1}{1+k^2\pi^2}\left(2 - 2\mathrm{e}^{-1}(-1)^k\right) \\
&= \frac{1}{1+k^2\pi^2}\left(1 - \mathrm{e}^{-1}(-1)^k\right).
\end{aligned}$$

As in a), we need to compute the integral of g(x) squared

$$\int_{-1}^{1} g^2(x)\mathrm{d}x = \int_{-1}^{1} e^{-2|x|}\mathrm{d}x = 1 - e^{-2}$$

$$E_N = 1 - e^{-2} - 2\sum_{k=-N}^{N}\left(\frac{1}{1+k^2\pi^2}\left(1 - \mathrm{e}^{-1}(-1)^k\right)\right)^2$$

Inserting values for N gives: $E_2 = 1.19e-3, E_4 = 2e-4, E_8 = 2.8e-5$

d) $g(x)$ is smoother than f(x) and the Fourier series for $g(x)$ are therefore a better approximation.

Problem 4. (Gibbs phenomenon)

In this exercise, we will illustrate what happens when we try to use Fourier series to describe a discontinuous function $f(x)$. Let's consider $1 - x^2$, $x \in [0, 1]$, and its odd extension to the interval $[-1, 0]$, $f(x) = sign(x)(1-x^2)$ which has a discontinuity at $x = 0$ and $L = 1$. As usual, the truncated trigonometric series can be written as

$$S_N(x) = a_0 + \sum_{n=1}^{N} a_n \cos n\pi x + b_n \sin n\pi x\,,\ \ n \in \mathbb{N}\,.$$

We will numerically investigate what happens to $S_N(x)$ when N is chosen larger and larger.

a) Determine an expression for the trigonometric Fourier coefficients a_0, a_n and b_n.

b) Using the Jupyter notebook `Gibbs.ipynb`, plot $f(x)$ and $S_N(x)$ for $N = 5, 20$ and 100.

c) We can easily verify that, at $x = 0$, the function's value jumps from -1 to 1. Let's define the global error as

$$\varepsilon_N := \max\{|S_N(x) - f(x)|\}\,,\quad x \in [-1, 1]\,,$$

as already implemented in the same Jupyter notebook. Following that, compute ε_N for $N = 1000$. In percentual terms, how large is this error with respect to the jump height 2?

Solution.

a) Being the function odd we can already conclude $a_0 = a_n = 0$. What remain is to compute the coefficients b_n:

$$b_n = \frac{2}{1}\int_0^1 (1 - x^2)\sin(n\pi x)\mathrm{d}x = 2\int_0^1 \sin(n\pi x)\mathrm{d}x - 2\int_0^1 x^2 \sin(n\pi x)\mathrm{d}x$$

We split it to obtain

$$2\int_0^1 \sin(n\pi x)\mathrm{d}x = -\frac{2}{n\pi}[\cos(n\pi) - 1] = 2\frac{1-(-1)^n}{n\pi}$$

and

$$\begin{aligned}\int_0^1 x^2\sin(n\pi x)\mathrm{d}x &= -\frac{1}{n\pi}[\cos(n\pi) - 0] + \frac{2}{n\pi}\int_0^1 x\cos(n\pi x)\mathrm{d}x \\ &= -\frac{(-1)^n}{n\pi} + \frac{2}{n\pi}\left[\frac{1}{n\pi}(\sin(n\pi) - 0) - \frac{1}{n\pi}\int_0^1 \sin(n\pi x)\mathrm{d}x\right] \\ &= -\frac{(-1)^n}{n\pi} - \frac{2}{n^2\pi^2}\int_0^1 \sin(n\pi x)\mathrm{d}x = -\frac{(-1)^n}{n\pi} + \frac{2}{n^3\pi^3}(\cos(n\pi) - 1) \\ &= -\frac{(-1)^n}{n\pi} + \frac{2(-1)^n}{n^3\pi^3} - \frac{2}{n^3\pi^3}\end{aligned}$$

Putting all together we have

$$b_n = \frac{2}{n\pi} - 4\frac{(-1)^n}{n^3\pi^3} + \frac{4}{n^3\pi^3}$$

b) See `Gibbs-complete.ipynb`.

c) Using again `Gibbs-complete.ipynb`, we compute $\varepsilon_{1000} \approx 0.17898$, which is around 8.95% of the jump height 2.

The next exercises are optional and should not be handed in!

Problem 5. (Convolution)

a) Compute the complex Fourier series of the 2π-periodic function

$$f(x) = \sin(3x) + 5\cos(2x).$$

b) Compute the complex Fourier series of the 2π-periodic function

$$g(x) = x^2, \quad -\pi \le x \le \pi,$$

c) Show that the convolution $h(x) = f(x) * g(x)$ is given as

$$h(x) = -\frac{4\pi}{9}\sin(3x) + 5\pi\cos(2x).$$

d) Compute the complex Fourier series of $h(x)$.

Hint. You can reuse the results from a) and b) and you do not have to solve integrals.

Solution.

a) The real Fourier coefficients are given as $a_2 = 5, b_3 = 1$ and all other coefficients are zero. This yields the complex coefficients $c_2 = c_{-2} = 5/2$, $c_3 = -\frac{1}{2}\mathrm{i} = \overline{c_{-3}}$ (since f is real) and all others are zero.

b) We can compute

$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 \,\mathrm{d}x = \frac{\pi^2}{3}.$$

as well as for $n \in \mathbb{N}$ the coefficients c_n, $n \in \mathbb{N}$ using two integraation by parts

$$\begin{aligned}
c_k &= \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 e^{-ikx}\,\mathrm{d}x\\
&= \frac{1}{2\pi}x^2\frac{e^{-ikx}}{-ik}\Big|_{-\pi}^{\pi} + \frac{1}{2\pi}\int_{-\pi}^{\pi} 2x\frac{e^{-ikx}}{ik}\,\mathrm{d}x\\
&= \frac{1}{2\pi}2x\frac{e^{-ikx}}{k^2}\Big|_{-\pi}^{\pi} - \frac{1}{2\pi}\int_{-\pi}^{\pi} 2\frac{e^{-ikx}}{k^2}\,\mathrm{d}x\\
&= \frac{2(-1)^k}{k^2} + \frac{1}{2\pi}\left(2\frac{e^{-ikx}}{ik^3}\Big|_{-\pi}^{\pi}\right)\\
&= \frac{2(-1)^k}{k^2}.
\end{aligned}$$

c) We first have to be careful about our integration interval in oder to not have to compute a modulus of g somewhere. Since the convolution is symmetric we can also just start with g and we are fine off (though the sine/cosine terms are a little bit more messy). So we start with

$$(f*g)(x) = (g*f)(x) = \int_{-\pi}^{\pi} g(y)f(x-y)\,\mathrm{d}y = \int_{-\pi}^{\pi} y^2\Big(\sin\big(3(x-y)\big)+5\cos\big(2(x-y)\big)\Big)\,\mathrm{d}y$$

We now use integration by parts. To be complete in $\int h_1h_2' = h_1h_2 - \int h_1'h_2$ we of course choose $h_1(y) = y^2$ since $h_1'(y) = 2y$ simplifies. On the other hand $h_2'(y) = \sin\big(3(x-y)\big)+5\cos\big(2(x-y)\big)$ can easily be integrated, we just have to keep in mind that we have an inner derivative from the summand $-3y$ (and $-2y$ resp.). We obtain with

$$h_2(y) = \frac{-\cos\big(3(x-y)\big)}{3} + \frac{5}{2}\sin\big(2(x-y)\big) = -\frac{1}{3}\cos\big(3(x-y)\big)+\frac{5}{2}\sin\big(2(x-y)\big)$$

That

$$\begin{aligned}
g*f(x) = y^2\Big(-\frac{1}{3}\cos\big(3(x-y)\big)+\frac{5}{2}\sin\big(2(x-y)\big)\Big)\Big|_{y=-\pi}^{\pi}\\
-\int_{-\pi}^{\pi} 2y\Big(-\frac{1}{3}\cos\big(3(x-y)\big)+\frac{5}{2}\sin\big(2(x-y)\big)\Big)\,\mathrm{d}y
\end{aligned}$$

Let's consider the first term. We first observe that for some a, $\sin(a\pm 2\pi) = \sin a\cos(2\pi)\pm\cos(a)\sin(2\pi) = \sin(a)$, since $\cos(2\pi) = 1$ and $\sin(2\pi) = 0$. With a similar reasoning

$\cos(3(x-\pi)) = \cos 3x \cos(3\pi)+\sin(3x)\sin(3\pi) = -\cos(3x)$. Hence the term simplifies to

$$\begin{aligned}\pi^2\Big(-\frac{1}{3}\cos\big(3(x-\pi)\big)+\frac{5}{2}\sin\big(2(x-\pi)\big)\Big) &- \pi^2\Big(-\frac{1}{3}\cos\big(3(x+\pi)\big)+\frac{5}{2}\sin\big(2(x+\pi)\big)\Big)\\ &= \pi^2\big(\frac{1}{3}\cos(3x)+\frac{5}{2}\sin(2x)-\frac{1}{3}\cos(3x)-\frac{5}{2}\sin(2x)\big) = 0\end{aligned}$$

For the integral we again do an integration by parts with (moving the $-$ into the integral) $h_1(y) = -2y$ so $h_1'(y) = -2$ and $h_2'(y) = -\frac{1}{3}\cos\big(3(x-y)\big)+\frac{5}{2}\sin\big(2(x-y)\big)$ so the antiderivative here reads

$$h_2(y) = -\frac{1}{9}\sin\big(3(x-y)\big) - \frac{5}{4}\cos\big(2(x-y)\big)$$

and we obtain

$$\begin{aligned}(g*f)(x) &= -2y\Big(-\frac{1}{9}\sin\big(3(x-y)\big)-\frac{5}{4}\cos\big(2(x-y)\big)\Big)\Big|_{y=-\pi}^{\pi}\\ &\qquad - \int_{-\pi}^{\pi}(-2)\Big(-\frac{1}{9}\sin\big(3(x-y)\big)-\frac{5}{4}\cos\big(2(x-y)\big)\Big)\,\mathrm{d}y\end{aligned}$$

Here we first start with the last term – the integral. Since we integrate (for fixed x) over 3 (2 for the second) whole periods, the integral is zero. For the first we have to plug in values for y

$$(g*f)(x) = -2\pi\Big(-\frac{1}{9}\sin\big(3(x-\pi)\big)-\frac{5}{4}\cos\big(2(x-\pi)\big)\Big)-(-2(-\pi))\Big(-\frac{1}{9}\sin\big(3(x+\pi)\big)-\frac{5}{4}\cos\big(2(x+\pi)\big)\Big)$$

Similar to the arguments above $\sin(3x \pm 3\pi) = -\sin(3x)$ and for some a we get $\cos(a \pm 2\pi) = \cos(a)\cos(2\pi) \mp \sin(a)\sin(2\pi) = \cos(a)$. We obtain

$$(f*g)(x) = -4\pi\Big(\frac{1}{9}\sin(3x)-\frac{5}{4}\cos(2x)\Big) = -\frac{4\pi}{9}\sin(3x)+5\pi\cos(2x).$$

d) The real Fourier coefficients are given as $a_2 = 5\pi$ and $a_n = 0$ for $a \neq 2$, moreover $b_3 = -\frac{4\pi}{9}$ and $b_n = 0$ for $n \neq 3$. This gives the complex coefficients $c_2 = \frac{5\pi}{2} = c_{-2}$ and $c_3 = i\frac{2\pi}{9}$, $c_{-3} = -i\frac{2\pi}{9}$. The Fourier coefficients of $f(x) * g(x)$ are given as 2π times the product of the Fourier coefficients of $f(x), g(x)$.

Problem 6. (Infinite sums)

In this exercise, we will use Fourier series and coefficients to show a few identities.

a) Consider the 2π-periodic function $f(x) = \begin{cases} -\pi - x & \text{if } -\pi < x < -\frac{\pi}{2}, \\ x & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ \pi - x & \text{if } \frac{\pi}{2} < x \leq \pi. \end{cases}$

Compute its Fourier coefficients and use the result, together with Parseval's identity, to show that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}\,.$$

b) Consider the function $f(x) = x$, defined for $x \in [0, 1]$. Compute the Fourier series of its *even extension*, and use the result to show that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \ldots = \frac{\pi^2}{8}\,.$$

c) Now, considering the same function $f(x)$ from (b), compute the Fourier series of its *odd extension* and use the result to show that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \ldots = \frac{\pi}{4}\,.$$

Solution.

a) We see that this is an odd function, meaning that $a_n = 0$.

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\mathrm{d}x = \frac{2}{\pi}\left(\int_0^{\pi/2} x\sin(nx)\mathrm{d}x + \int_{\pi/2}^{\pi}(\pi - x)\sin(nx)\mathrm{d}x\right)$$
$$= \begin{cases} \frac{4(-1)^{n+1}}{\pi n^2} & \text{if n is odd} \\ 0 \quad \text{if n is even} \end{cases}$$

Now integrating f^2

$$\int_0^{\pi} f^2(x)\mathrm{d}x = \frac{\pi^3}{6} \tag{5}$$

and using Parseval's identity gives,

$$\frac{\pi^3}{6} - \sum_{n=1}^{N} \frac{16}{\pi(2n+1)^4} = 0$$

$$\sum_{n=1}^{N} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}$$

b) If we reuse our computations from 2d) we have that

$$a_n = \frac{2((-1)^n - 1)}{n^2\pi^2} = \begin{cases} \frac{-4}{\pi n^2} & \text{if n is odd} \\ 0 & \text{if n is even} \end{cases}$$

In addition, we need to compute a_0,

$$a_0 = \int_0^1 x \mathrm{d}x = \frac{1}{2}$$

The Fourier series for the function can therefore be written as

$$f(x) = \frac{1}{2} - \frac{4}{\pi^2}\left(\frac{\cos(\pi x)}{1^2} + \frac{3\cos(\pi x)}{3^2} + \frac{5\cos(\pi x)}{5^2} + \dots\right)$$

and by setting $x = 0$ we have that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \dots = \frac{\pi^2}{8}$$

c) Now, we can reuse our computations from 1c)

$$b_n = \frac{2(-1)^{n+1}}{n\pi},$$

and the Fourier series can be written as

$$f(x) = \frac{2}{\pi}\left(\frac{\sin(\pi x)}{1} - \frac{\sin(2\pi x)}{2} + \frac{\sin(3\pi x)}{3} - \frac{\sin(4\pi x)}{4} + \frac{\sin(5\pi x)}{5} + \dots\right).$$

By setting $x = \frac{\pi}{2}$ we have that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots = \frac{\pi}{4}.$$

Problem 7. (Derivative)
Let $c_k(f)$ denote the complex Fourier coefficients for the twice continuously differentiable and 2π-periodic function $f(x)$. Show that the complex Fourier coefficients $c_k(f')$ for $f'(x)$ are given by $c_k(f') = ikc_k$.

Solution.
Recall that the c_k are given by

$$c_k(f) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}dx.$$

Then integration by parts gives:

$$\begin{aligned}
c_k(f') =& \frac{1}{2\pi}\int_{-\pi}^{\pi} f'(x)e^{-ikx}dx \\
=& \frac{1}{2\pi}([f(x)e^{-ikx}]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} -ik \cdot f(x)e^{-ikx}dx) \\
=& ikc_k + \frac{1}{2\pi}(f(\pi)e^{-ik\pi} - f(-\pi)e^{ik\pi}) \\
=& ikc_k + \frac{1}{2\pi}(f(\pi)e^{ik\pi} - f(\pi)e^{ik\pi}) \\
=& ikc_k,
\end{aligned}$$

where we use that $f(x)$ is 2π-periodic and that $e^{ik\pi} = e^{-ik\pi}$ for integer values of k.